Round 10 | Can the young tier be halved?

One mechanism, an exact cost obstruction, and a stored-leg test | 24 September 2026

Answer: no qualifying method established. I tested a different arithmetic representation: **two-way numerical packing in float64**. It retains the full dense legs and realized signs. It can combine two products into one, but the installed production meter charges float64 at twice float32. Thus this mechanism cannot deliver the requested 104 u to 55 u reduction. This is a no-go result for this packing substitution, **not for every possible representation**.

The one method assessed

Keep the original P and A for every young source; four ages occupy $8n^2$ float32 numbers, or 32 MiB. Form quantized integer-valued operands in float64, plus $O(n^2)$ packing buffers. No source, hub, tensor entry, or sign is averaged away. Numerical packing is an established approximate matrix-multiplication technique [1]; the following application is the method assessed here.

For transport, quantize M, P and A with known scales, and calculate one square product Mq(Pq + B Aq). Decode its low and high signed radix-B coefficients to obtain both transported legs, then undo their individual scales. Here M is the actual transport map, including the Wick scaling when applicable.

For the requested slice, define four n-by-n factors (the division by 3 is included):

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L1 = (P*P) diag(w)/3; R1 = A.T
L2 = 2(P*A) diag(w)/3; R2 = P.T
D21 = L1 R1 + L2 R2. (* means entrywise product.)
```

Choose quantization scales so both desired products have the same dequantization factor s. For the integer-valued quantized factors, compute

```
For reproducibility, set Q = 511, c = max over j of max|Lj| max|Rj|,
aj = max|Lj|/Q, bj = c/(Q^2 aj), Ljq = round(Lj/aj),
Rjq = round(Rj/bj), and s = c/Q^2. Zero terms can be skipped.
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Z = (B L1q + L2q)(B R2q + R1q)
= B^2 L1q R2q + B(L1q R1q + L2q R2q) + L2q R1q.
T = round(Z/B); Dhat = s [T - B round(T/B)].
```

In exact arithmetic this extracts the middle coefficient when the low coefficient and the signed middle coefficient lie strictly between -B/2 and B/2. Floating-point error also needs sufficient guard space. The state can reproduce odd fluctuations because it encodes the actual signed legs. **Quantization and Strassen roundoff still require validation; algebra alone does not guarantee 2% accuracy.**

Exact cost obstruction

Let S be the charged cost of one float32 square product at the chosen Strassen depth. The same arithmetic in float64 costs 2S. A pair of float32 products therefore costs 2S before substitution and 2S after substitution, before packing, quantization and decoding. Those additional operations have nonnegative cost. Shared left-operand preprocessing in the original transport does not create an advantage for the packed version.

Consequently, replacing every eligible two-product pair cannot halve its product bill. At the prompt's S approximately 0.51 u, four ordinary products cost 2.04 u; two packed float64 products also cost 2.04 u, plus preprocessing. Any unpaired newborn product remains chargeable.

Exact shape ledger and the falsifier

All dense products below are (1024 x 1024) by (1024 x 1024).

At a layer with all four young ages, the age-1 P = W is already available. Three older young sources need six transports; the newborn A needs one. Four sources need eight slice products. Covariance riders are outside this source-only count and remain charged.

Source-only operation	Original	Two-way packed
Ages 2-4 transport	6 float32 squares	3 float64 squares
Age-1 A transport	1 float32 square	1 float32 square
Four D21 contributions	8 float32 squares	4 float64 squares
Total, before elementwise work	15S	7(2S) + S = 15S

For the 16-layer schedule, with the first birth read at layer 2 and the shipped final-layer d3-only shortcut, the square-product counts are 0, 3, 7, 11, then 15 at layers 5-15, and 7 at layer 16: **193 float32-product equivalents in either representation**. At 0.51 u each this is 98.43 u before the separately charged work. This ideal source-only ledger is not an exact reconstruction of the measured approximately 104 u family ledger, whose covariance riders and other allocations must be reconciled. The equality of the before/after product bill does not depend on that allocation.

For a conventional five-level recursion with 32-by-32 leaves, the exact leaf arithmetic is $7^5(2 \times 32^3 - 32^2)$ = **1,084,253,184** operations per square. Standard Strassen adds 96,970,752 block additions/subtractions. These and packing operations must also be charged; counting a float64 product as 0.51 u would omit its dtype multiplier.

One falsifier, executed on a stored source

Use **data/dumps/young_dump_L14.npz, source index 12**, and the two-term slice formula in the prompt. Run two ordinary float32 square products and one packed float64 square product under the installed meter. Stop if the product-only cost ratio exceeds 55/104 = 0.52885; adding packing overhead cannot repair that failure.

Readout	Measured result
Environment	flopscope 0.12.1; whestbench 0.16.1; no weight overrides
Two ordinary float32 matmuls	4,292,870,144 charged FLOPs
One packed float64 matmul	4,292,870,144 charged FLOPs
Product-only ratio / cost gate	1.0000 / FAIL
Q = 511; B = 2^{31}	D21 error 1.1858%; diagonal error 0.6495%
Decoded vs. separately multiplied quantized reference	0 difference in this run

The reference for these errors uses the original stored legs in float64. The test uses **classical matmul on one source**, not Strassen, not the full fused production slice, and not a recurrent four-age rollout. Quantization accounts for the observed error. Quantization, packing and decoding were excluded from the charged totals to give the candidate a favorable lower bound; its full bill is higher.

Conclusion: this concrete mechanism fails the cost gate even on a slice where its numerical error is below 2%. I have no justified 55 u method and no general arithmetic lower bound ruling one out. This result establishes only that two float32 products replaced by one float64 product cannot supply the required savings under this meter.

[1] Anastasia & Andreopoulos, Throughput-Distortion Computation of Generic Matrix Multiplication, IEEE Transactions on Signal Processing (2012), author manuscript. Cited for numerical packing, not for ARC accuracy or billing.
Local billing evidence: installed flopscope _dtype_billing.py / _weights.py and the executed BudgetContext probe. Accuracy target and network ledger: supplied Round 10 prompt. No estimator or repository source was edited.